

MANUEL PAEZ

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EDUCATION

Columbia University

New York, NY

Master of Science, Computer Science, GPA: 3.8/4.0

Jan. 2025 - Expected Dec 2026

Graduate Coursework: Advanced Probability Theory, Computational Learning Theory, Probability II, Projects in CS, Statistical Inference, Theoretical Foundations of LLMs.

Columbia University

New York, NY

Bachelor of Arts, Computer Science and Mathematics

Sept. 2023 - May 2026

Bachelor of Arts, Computer Science

Sept. 2019 - May 2023

Relevant Coursework: Advanced Algorithms, Advanced Topics in Deep Learning, Algorithms for Massive Data, Analysis and Probability I, Honors Complex Variables, Machine Learning Theory, Natural Artificial Neural Networks, Real Analysis I and II, Theoretical Neuroscience, Unsupervised Learning, Intro to Quantum Computing.

RESEARCH AND WORK EXPERIENCE

Columbia University, Department of Computer Science

New York, NY

Researcher, supervised by Professor Alexandr Andoni

Jan. 2024 - Present

- Developed randomized constructions for high-dimensional Euclidean and Hamming spanners using batch approximate nearest-neighbor search and locality-sensitive hashing.
- Studied low-distortion embeddings of disagreement metrics from tree-structured Bayesian networks and Markov random fields into finite-dimensional ℓ_1 spaces.
- Contributed to theoretical analyses of approximation guarantees, graph sparsity, and algorithmic runtime.

Flatiron Institute, Simons Foundation

New York, NY

Neural Circuits and Algorithms Group, Center for Computational Neuroscience

Visiting Researcher

Feb. 2026 - Present

- Developed biologically plausible algorithms for learning low-dimensional predictive representations of nonlinear dynamical systems using rectified and asymmetric similarity matching.
- Implemented past-future representation-learning methods in Python and evaluated them on neuronal and nonlinear dynamical-system trajectories.

Visiting Scholar Fellow

Nov. 2024 - Aug. 2025

- Completed the migration and integration of CaImAn's deep-learning components from TensorFlow to PyTorch.
- Supported model validation and scalable calcium- and voltage-imaging workflows using Flatiron Institute's high-performance computing infrastructure.

Summer Research Intern

May 2024 - Aug. 2024

- Migrated deep-learning components of CaImAn's calcium- and voltage-imaging pipelines from TensorFlow to PyTorch.
- Reimplemented an automated and scalable voltage-imaging analysis pipeline with a PyTorch backend.
- Trained and evaluated supervised models on more than 10 TB of imaging data using Slurm-based HPC workflows.
- Collaborated with scientists at HHMI Janelia on imaging-analysis development and validation.

Research Intern, Neural Circuits and Algorithms Group

Sept. 2022 - Aug. 2023

- Developed and evaluated an IsoRSUNet-based model for neuron-boundary segmentation in large-scale connectomic electron-microscopy volumes.
- Conducted ablation studies to assess segmentation accuracy, architectural stability, and generalization.
- Built HPC workflows for automated model training, inference, and segmentation using PyTorch Connectomics and Neuroglancer.

Columbia University, Department of Physics

New York, NY

Undergraduate Researcher, supervised by Professor Szabolcs Márka

Oct. 2022 - May 2023

- Implemented a quantum pattern-recognition algorithm in Python and Qiskit for detecting candidate signals in gravitational-wave data.
- Developed evaluation methods using publicly available LIGO data and collaborated with researchers at Heidelberg University on algorithm design and validation.

Columbia University, Department of Psychiatry

New York, NY

Undergraduate Researcher, supervised by Professor Kiyohito Iigaya

April 2022 - Sept. 2022

- Analyzed neural representations from context-dependent tasks using principal component analysis and manifold-learning methods.
- Built modified LSTM models in PyTorch and TensorFlow to reconstruct latent neural geometry from primate behavioral and neural recordings, in collaboration with researchers at the University of Cambridge.

PUBLICATIONS

Alexandr Andoni, Manuel Paez, Krish Singal, and Tian Zhang. “Fast Spanners and Clustering via Batch Approximate Nearest Neighbor Search.” Submitted manuscript, 2026.

Manuel Paez. “Task-Response and Curvature Spectral Geometry of High-Dimensional Optimization.” Submitted manuscript, 2026.

Manuel Paez and Chryseis Xinyi Liu. “Rank-One Spectral Normalization Accelerates Projected Feature Learning.” ICML 2026 Workshop on High-Dimensional Learning Dynamics, 2026. [\[Paper\]](#) [\[Code\]](#)

PRESENTATIONS

Manuel Paez and Chryseis Xinyi Liu. “Rank-One Spectral Normalization Accelerates Projected Feature Learning.” Presented at the ICML 2026 Workshop on High-Dimensional Learning Dynamics, Vancouver, Canada, July 2026.

Manuel Paez. “CaImAn PyTorch: Calcium Image Data Analysis Tool for PyTorch.” Simons Foundation Summer Intern Research Symposium, New York, NY, August 2024.

Manuel Paez. “Self-supervised Neuron Boundary Inpainting to Fix Membrane Leaking in Electron Microscopy Images.” Poster presented at the Simons Foundation SURFiN Research Symposium, New York, New York, April 2023.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, R, Java

Machine Learning: pandas, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, SciPy, Jupyter

Systems and Scientific Computing: Linux, Git, Slurm and HPC computing, scikit-learn, VS Studio Code, Vim.

Specialized Tools: Qiskit, Cirq, LaTeX, Neuroglancer

Natural Languages: English (Fluent), Spanish (Fluent), German (Professional working proficiency), French (Elementary), Korean (Elementary)

TEACHING EXPERIENCE

Columbia University

New York, NY

Course Assistant, Department of Computer Science

Sept. 2024 - Dec. 2025

- Taught Recitations, held office hours, and graded assignments for COMS 4995: Randomized Algorithms and COMS 6998: Algorithms for Massive Data.

AWARDS

Dean’s List

Summer 2021, Fall 2022, Spring 2024 - Spring 2025

Climate Science Olympiad 2024, Semi-finalist

Aug. 2024

MIT IQuHACK Hackathon 2023; Covalent x IBM Challenge - 1st place

Jan. 2023

Simons Foundation Global Brain SURF Fellowship
United States International Young Physicist Tournament (USIYPT) - 1st place

Aug. 2022
Feb. 2018 and Jan. 2019

LEADERSHIP AND EXTRACURRICULAR EXPERIENCE

Columbia Undergraduate Quantum Computing Club (CUQCC)

Dec. 2022 - May 2023

Co-head and Co-founder

- Organized seminars and technical events featuring speakers from students, faculty, and research scientists across quantum information science.
- Managed the club website and weekly member communications.
- Led the team to first place in the IBM x Covalent Challenge MIT IQuHACK 2023 for a quantum-chaos approach to energy-grid optimization.

Markov Chains and Mixing Times Reading Group

May 2025 - Aug. 2025

Columbia Undergraduate Theoretical Computer Science Learning Seminar

Sept. 2023 - May 2026

Undergraduate Math Society (UMS)

Sept. 2019 - Dec. 2024

Supervised Undergraduate Mathematics Reading Group - Combinatorics

Sept. 2023 - Dec. 2023